MATH3900

Tutorial 4

1. Let $f : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable, μ-strongly convex and L-smooth, where $0 < \mu \le L$. Consider the following algorithm with zero stopping tolerance.

Newton's method with Armijo backtracking

Input: $x_0 \in \mathbb{R}^d$, $\bar{\alpha} > 0$, and $c_A, \eta \in (0, 1)$.

for $k = 0, 1, 2, \ldots$ **do**
  Compute $g_k = \nabla f(x_k)$.
  if $g_k = 0$ **then**
    stop and return x_k.
  end if
  Compute $H_k = \nabla^2 f(x_k)$; solve $H_k p_k = -g_k$.
  Set $\alpha = \bar{\alpha}$.
  while $f(x_k + \alpha p_k) > f(x_k) + c_A \alpha g_k^T p_k$ **do**
    Set $\alpha = \eta\alpha$.
  end while
  Set $\alpha_k = \alpha$ and $x_{k+1} = x_k + \alpha_k p_k$.
end for

You may use the Hessian bounds $\mu I \preceq H_k \preceq LI$. The global convergence theorem proved in class may be used without proof: for a bounded-below, L-smooth objective, the uniform bounds

$$g_k^T p_k \le -\sigma \|g_k\|^2, \qquad \|p_k\| \le \beta \|g_k\|$$

with fixed $\sigma, \beta > 0$ imply $\|g_k\| \to 0$.

Prove that this algorithm converges to the unique minimizer $x^\star$ from any starting point, and that the additional assumptions in part (d) give eventual quadratic convergence.

(a) **Uniform direction bounds.** Show that Newton's direction satisfies the theorem's bounds with $\sigma = 1/L$ and $\beta = 1/\mu$.

(b) **Distance to the minimizer.** Using strong convexity at x and $x^\star$, show that

$$\|x - x^\star\| \le \frac{\|\nabla f(x)\|}{\mu}.$$

(c) **Global convergence.** Apply the lecture's theorem and part (b) to prove that Newton with backtracking converges to $x^\star$ from any starting point.

(d) **Eventual full steps.** Suppose additionally that $\bar{\alpha} = 1$, $c_A < 1/2$, and the Hessian is ρ-Lipschitz, with $\rho \ge 0$, in a neighborhood of $x^\star$:

$$\left\|\nabla^2 f(u) - \nabla^2 f(v)\right\|_2 \le \rho \|u - v\|$$

for u, v in that neighborhood. At a point x, write $g = \nabla f(x)$, $H = \nabla^2 f(x)$, and let $Hp = -g$. If the full-step segment stays in this neighborhood, you may use

$$f(x+p) \le f(x) + g^T p + \tfrac{1}{2} p^T H p + \tfrac{\rho}{6} \|p\|^3 .$$

Show that the full step satisfies Armijo whenever

$$\tfrac{\rho}{6} \|p\| \le \mu(1/2 - c_A).$$

Deduce that the algorithm eventually accepts $\alpha_k = 1$.

(e) **Quadratic convergence.** Under the additional assumptions in part (d), prove that, for all sufficiently large k,

$$\|x_{k+1} - x^\star\| \le \frac{\rho}{2\mu} \|x_k - x^\star\|^2 .$$

Explain what this inequality says about the error in successive iterations. *Hint:* with $e = x - x^\star$ and $H(z) = \nabla^2 f(z)$, the fundamental theorem of calculus gives

$$\nabla f(x) = \int_0^1 H(x^\star + te) e \, dt.$$

Solution: (a) Uniform direction bounds. The eigenvalues of H_k lie in $[\mu, L]$. They are positive, so H_k is invertible, and its inverse has eigenvalues in $[1/L, 1/\mu]$. Hence

$$\frac{1}{L} I \preceq H_k^{-1} \preceq \frac{1}{\mu} I.$$

Substitute $p_k = -H_k^{-1} g_k$:

$$g_k^T p_k = -g_k^T H_k^{-1} g_k \leq -\frac{1}{L} \|g_k\|^2 .$$

The inequality reverses when the quadratic form is multiplied by -1. Also, since $\left\|H_k^{-1}\right\|_2 \leq 1/\mu$,

$$\|p_k\| = \left\|H_k^{-1} g_k\right\| \leq \left\|H_k^{-1}\right\|_2 \|g_k\| \leq \frac{1}{\mu} \|g_k\| .$$

Thus $\sigma = 1/L$ and $\beta = 1/\mu$ work at every iteration. In particular, $g_k^T p_k < 0$ whenever $g_k \neq 0$.

(b) Distance to the minimizer. Write $g = \nabla f(x)$ and $e = x - x^\star$. Since $x^\star$ minimizes a differentiable unconstrained objective, $\nabla f(x^\star) = 0$. Strong convexity in the two directions gives

$$f(x^\star) \geq f(x) + g^T (x^\star - x) + \tfrac{\mu}{2} \|e\|^2 ,$$
$$f(x) \geq f(x^\star) + 0^T (x - x^\star) + \tfrac{\mu}{2} \|e\|^2 .$$

Add the inequalities and cancel the function values:

$$0 \geq -g^T e + \mu \|e\|^2 \quad \Longrightarrow \quad \mu \|e\|^2 \leq g^T e \leq \|g\| \|e\| .$$

The last step is Cauchy–Schwarz. If $e \neq 0$, divide by $\mu \|e\| > 0$ to obtain $\|e\| \leq \|g\| / \mu$. If $e = 0$, the same conclusion is immediate.

(c) Global convergence. Strong convexity supplies a quadratic lower bound:

$$f(x) \geq f(0) - \|\nabla f(0)\| \|x\| + \tfrac{\mu}{2} \|x\|^2 .$$

This is bounded below and tends to $+\infty$ as $\|x\| \to \infty$. Continuity therefore gives a minimizer, and strong convexity makes it unique. Thus the bounded-below hypothesis of the lecture's theorem is satisfied.

The objective is L-smooth, and part (a) supplies fixed positive direction constants. The theorem yields $\|g_k\| \to 0$. Part (b) now gives

$$0 \leq \|x_k - x^\star\| \leq \frac{\|g_k\|}{\mu} \longrightarrow 0.$$

Hence $x_k \to x^\star$. If a zero gradient is reached in finitely many iterations, part (b) shows that this point is already $x^\star$, and the method stops.

Solution: (d) Eventual full steps. The Newton equation gives $g^T p = -p^T H p$. Substitute this into the supplied Taylor bound:

$$\begin{aligned} f(x+p) &\le f(x) - p^T H p + \tfrac{1}{2} p^T H p + \tfrac{\rho}{6} \|p\|^3 \\ &= f(x) - \tfrac{1}{2} p^T H p + \tfrac{\rho}{6} \|p\|^3 . \end{aligned}$$

For $\alpha = 1$, the Armijo bound is $f(x) + c_A g^T p = f(x) - c_A p^T H p$. Subtract that bound from the Taylor upper bound:

$$\begin{aligned} f(x+p) - [f(x) + c_A g^T p] &\le -(1/2 - c_A) p^T H p + \tfrac{\rho}{6} \|p\|^3 \\ &\le \|p\|^2 [-\mu(1/2 - c_A) + \tfrac{\rho}{6} \|p\|]. \end{aligned}$$

Here $p^T H p \ge \mu \|p\|^2$ and $1/2 - c_A > 0$. The last expression is nonpositive under the stated condition, so the full step passes Armijo.

Part (c) gives $x_k \to x^\star$, and part (a) gives $\|p_k\| \le \|g_k\| / \mu \to 0$. Choose a small ball around $x^\star$ within the Hessian-Lipschitz neighborhood. Eventually both x_k and the whole segment $x_k + t p_k$, $0 \le t \le 1$, lie in this ball. Also $(\rho/6) \|p_k\| \to 0$, whereas $\mu(1/2 - c_A) > 0$. The full-step condition therefore holds for all sufficiently large k. Since backtracking starts with $\bar{\alpha} = 1$, it then accepts 1 at the first trial.

Solution: (e) Quadratic convergence.

From that point onward the method is ordinary full-step Newton near $x^\star$. To prove the quadratic error bound, write $H(z) = \nabla^2 f(z)$, $g = \nabla f(x)$, and $e = x - x^\star$. Since $\nabla f(x^\star) = 0$, the fundamental theorem of calculus gives

$$g = \int_0^1 H(x^\star + te)e \, dt.$$

A full Newton update satisfies

$$\begin{aligned} x + p - x^\star &= e - H(x)^{-1} g \\ &= H(x)^{-1}\big[H(x)e - g\big] \\ &= H(x)^{-1} \int_0^1 [H(x) - H(x^\star + te)]e \, dt. \end{aligned}$$

Here $\|H(x)^{-1}\|_2 \le 1/\mu$, and the Lipschitz condition gives

$$\|H(x) - H(x^\star + te)\|_2 \le \rho \, \|x - (x^\star + te)\| = \rho(1-t) \, \|e\| \, .$$

Taking norms and using these two bounds,

$$\begin{aligned} \|x + p - x^\star\| &\le \frac{1}{\mu} \int_0^1 \rho(1-t) \, \|e\|^2 \, dt \\ &= \frac{\rho}{2\mu} \, \|e\|^2 \, . \end{aligned}$$

Both segments used above are eventually inside the chosen ball. Applying this bound at $x = x_k$ gives the required inequality for all sufficiently large k. This is eventual quadratic convergence: once the iterates are close enough, the next error is at most a fixed constant times the current error squared. If $\rho = 0$, the remainder vanishes; the full-step test holds automatically there, and the error bound gives exact termination.

2. Determine whether each statement is *True* or *False*. Give a short proof for a true statement, or a counterexample with verification for a false statement.

   (a) For fixed c_A, backtracking always returns the largest positive step size satisfying Armijo.

   (b) If $f : \mathbb{R} \to \mathbb{R}$ is bounded below and has a Lipschitz continuous derivative, every point with $f'(t) = 0$ is a global minimizer.

Solution:

(a) False. Let $f(t) = t^2/2$, $t = 1$, $p = -1$, $c_A = 1/4$, $\bar{\alpha} = 1$, and $\eta = 1/2$. The first trial passes:

$$f(0) = 0 \le \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4}.$$

Backtracking returns 1, although $\alpha = 3/2$ also passes:

$$f(1 - \tfrac{3}{2}) = \tfrac{1}{8} = \tfrac{1}{2} - \tfrac{1}{4} \cdot \tfrac{3}{2}.$$

The method chooses the first passing trial in its prescribed sequence.

(b) False. Take $f(t) = 1 - \cos t$. It is bounded below by zero, and $f'(t) = \sin t$, $f''(t) = \cos t$. The mean value theorem and $|\cos t| \le 1$ show that $|f'(u) - f'(v)| \le |u - v|$, so the derivative is 1-Lipschitz. But $f'(\pi) = 0$ and $f(\pi) = 2 > f(0) = 0$. The general theorem gives $\|\nabla f(x_k)\| \to 0$; it does not say that every stationary point is a global minimizer. Question 1 uses strong convexity to obtain the stronger conclusion.

3. Let

$$f(x) = x_1^2 + 3x_2^2, \qquad x = (1,1)^T, \qquad p = -\nabla f(x).$$

Use Armijo backtracking with $\bar{\alpha} = 1$ and $\eta = 1/2$.

(a) Compute p, $\nabla f(x)^T p$, and $\phi(\alpha) = f(x + \alpha p)$.

(b) Take $c_A = 1/4$. Check the trials in order until the first acceptance. For each trial, give the objective value and the Armijo bound. Find the accepted point.

(c) Find the exact interval of acceptable positive steps for $c_A \in (0,1)$. Use it to find the first accepted trial when $c_A = 3/4$. Give a rejected trial that nevertheless decreases f.

Solution: **(a)** At the given point,

$$f(x) = 4, \quad g = (2,6)^T, \quad p = (-2,-6)^T, \quad g^T p = -40.$$

The trial point is $(1 - 2\alpha, 1 - 6\alpha)^T$. Expanding gives

$$\begin{aligned} \phi(\alpha) &= (1-2\alpha)^2 + 3(1-6\alpha)^2 \\ &= 1 - 4\alpha + 4\alpha^2 + 3 - 36\alpha + 108\alpha^2 \\ &= 4 - 40\alpha + 112\alpha^2. \end{aligned}$$

(b) With $c_A = 1/4$, the Armijo bound is $4 - 10\alpha$.

α	$\phi(\alpha)$	$4 - 10\alpha$	decision
1	$4 - 40 + 112 = 76$	-6	reject
1/2	$4 - 20 + 28 = 12$	-1	reject
1/4	$4 - 10 + 7 = 1$	3/2	accept

The first accepted step is 1/4, giving $x_{\text{new}} = (1,1)^T + \frac{1}{4}(-2,-6)^T = (1/2, -1/2)^T$. Its objective value is 1.

Solution: **(c)** Substitute ϕ into the Armijo inequality:

$$\begin{aligned} 4 - 40\alpha + 112\alpha^2 &\le 4 - 40c_A\alpha, \\ 112\alpha^2 &\le 40(1 - c_A)\alpha. \end{aligned}$$

Since $\alpha > 0$, divide by $112\alpha > 0$. The exact acceptable interval is

$$0 < \alpha \le \frac{5(1 - c_A)}{14}.$$

For $c_A = 3/4$, its upper endpoint is $5/56$. The trials $1, 1/2, 1/4, 1/8$ exceed this endpoint, whereas $1/16 < 5/56$. Thus $1/16$ is the first accepted trial, with

$$x_{\text{new}} = (1, 1)^T + \tfrac{1}{16}(-2, -6)^T = (7/8, 5/8)^T.$$

For example, the rejected trial $1/8$ gives

$$\phi(1/8) = 4 - 5 + \tfrac{7}{4} = \tfrac{3}{4} < 4, \qquad 4 - 40c_A(1/8) = 4 - \tfrac{30}{8} = \tfrac{1}{4}.$$

It decreases the objective, but $3/4 > 1/4$, so it fails Armijo.